

# Enterprise Network Design & Simulation

## Problem Statement and Proposed Solution

### 1. Problem Statement

Modern organizations contain multiple departments that need reliable network connectivity while maintaining logical separation of departmental traffic. A flat network can make management difficult, increase unnecessary broadcast traffic, and reduce control over communication between departments.

The problem is to design and simulate an enterprise network in which different departments are separated using Virtual Local Area Networks (VLANs), while still allowing communication between authorized networks through inter-VLAN routing. The network must provide appropriate IP addressing, trunk connections, access ports, and connectivity to a server network.

### 2. Project Objectives

|   |                                                       |
|---|-------------------------------------------------------|
| 1 | Design a scalable enterprise network topology.        |
| 2 | Separate departments using VLANs.                     |
| 3 | Assign IPv4 addresses and default gateways.           |
| 4 | Configure access and trunk ports on switches.         |
| 5 | Implement inter-VLAN routing using Router-on-a-Stick. |
| 6 | Provide connectivity to the server network.           |
| 7 | Test and verify connectivity using ping.              |

### 3. Proposed Network

The proposed solution uses Cisco Packet Tracer to simulate an enterprise network consisting of one Cisco 2911 router, four switches, six PCs, and one server. The departments are represented by separate VLANs. A Router-on-a-Stick configuration provides the gateway for each VLAN and enables inter-VLAN communication.

### 4. VLAN and IP Addressing Plan

| VLAN | Department | Network         | Gateway      |
|------|------------|-----------------|--------------|
| 10   | HR         | 192.168.10.0/24 | 192.168.10.1 |
| 20   | Finance    | 192.168.20.0/24 | 192.168.20.1 |
| 30   | IT         | 192.168.30.0/24 | 192.168.30.1 |
| 40   | Server     | 192.168.40.0/24 | 192.168.40.1 |

## 5. Main Network Requirements

- Cisco Packet Tracer simulation environment.
- Cisco 2911 router for inter-VLAN routing.
- Core switch and departmental switches.
- Separate VLANs for HR, Finance, IT, and Server networks.
- IPv4 /24 addressing for departmental networks.
- Trunk links between network devices where required.
- Access ports for end devices.
- Connectivity testing using ping.

## 6. Expected Outcome

The completed simulation should provide logical separation between departments using VLANs and allow communication between different VLANs through the router. The network should be easy to understand, manage, test, and demonstrate in Cisco Packet Tracer. Successful connectivity tests should verify the operation of the configured network.

## 7. Team

| Name             | Role                           |
|------------------|--------------------------------|
| Prapul Rajakumar | Team Lead / Network Design     |
| Praveen Wali     | Configuration & Implementation |
| Manvith Kumar N  | Testing & Documentation        |

## Conclusion

This project addresses the need for a structured enterprise network by combining VLAN-based departmental separation, IPv4 addressing, switching, trunking, and Router-on-a-Stick inter-VLAN routing in a Cisco Packet Tracer simulation.